Results

Sensitivity with rho = 0.4

Strength

Pooled meta-analysis for maximal strength identified a practically not worthwhile effect favouring slower eccentric duration (g = 0.18 [-0.38, 0.73], 90% CI: [-0.27, 0.62], 95%PI = [-1.23, 1.59], 90%PI [-0.95, 1.30], p = 0.48, I^2_{within} = 0%, $I^2_{between}$ = 65.49%, τ^2_{within} < 0.01, $\tau^2_{between}$ = 0.30).

Sub-group analysis identified a practically not worthwhile effect for longer eccentrics for between- (g = 0.21 [-0.69, 1.12], 90% CI: [-0.49, 0.92], 95%PI = [-1.92, 2.34], 90%PI = [-1.91, 2.33], p = 0.57, I^2_{within} = 0%, $I^2_{between}$ = 73.18%, τ^2_{within} < 0.001, $\tau^2_{between}$ = 0.56) and within- (g = 0.11 [-1.17, 1.39], 90% CI: [-0.52, 0.75], 95%PI = [-1.17, 1.39], 90%PI = [-0.52, 0.74], p = 0.47, I^2_{within} = 0%, $I^2_{between}$ = 0%, τ^2_{within} < 0.01, $\tau^2_{between}$ < 0.01) participant study designs. Moderator analysis did not suggest differences between between-participant and within-participant subgroups (β = -0.11 [-1.72, 1.51], 90% CI [-1.20, 0.99], F[1, 1.94] = 0.08, p = 0.80).

Sub-group analysis of training status identified a practically not worthwhile effect favouring shorter eccentric duration for untrained participants (g = -0.03 [-1.67, 1.60], 90% CI: [-1.24, 1.18], 95%PI = [-3.49, 3.42], 90%PI = [-2.58, 2.52], p = 0.95, I^2_{within} = 0%, $I^2_{between}$ = 84.51%, τ^2_{within} < 0.01, $\tau^2_{between}$ = 0.91). In trained participants, there was a small effect favouring longer eccentric durations (g = 0.34 [-0.04, 0.72], 90% CI: [0.07, 0.60], 95%PI = [-0.04, 0.72], 90%PI = [0.07, 0.60], p = 0.06, I^2_{within} = 0%, $I^2_{between}$ = 0%, τ^2_{within} < 0.01, $\tau^2_{between}$ < 0.01). Moderator analysis did not suggest differences between trained and untrained subgroups (β = -0.42 [-1.61, 0.77], 90% CI [-1.36, 0.52], F[1, 5.86] = 0.74, p = 0.42).

Sub-group analysis of multi-joint exercise selection identified a practically not worthwhile effect for shorter eccentric durations (g = -0.10 [-1.41, 1.20], 90% CI: [-1.07, 0.86], 95%PI = [-2.75, 2.54], 90%PI = [-2.06, 1.85], p = 0.81, I^2_{within} = 0%, $I^2_{between}$ = 73.21%, τ^2_{within} < 0.001, $\tau^2_{between}$ = 0.52). For single joint exercise selection, there was a small effect for longer eccentric durations (g = 0.40 [-0.47, 1.27], 90% CI: [-0.22, 1.02], 95%PI = [-1.10, 1.90], 90%PI = [-0.69, 1.48], p = 0.23, I^2_{within} = 24.88%, $I^2_{between}$ = 24.88%, τ^2_{within} = 0.06, $\tau^2_{between}$ = 0.06). Moderator analysis did not suggest differences between multi-joint and single-joint subgroups (β = 0.54 [-0.65, 1.73], 90% CI [-0.40, 1.48], F[1, 5.85] = 1.24, p = 0.31).

Sub-group analysis of volume matched identified a small effect for longer eccentric durations (g = 0.25 [-0.03, 0.52], 90%CI = [0.05, 0.44], 95%PI = [-0.05, 0.55], 90%PI = [0.05, 0.44], p = 0.06, I^2_{within} = 0%, $I^2_{between}$ = 0%, τ^2_{within} < 0.01, $\tau^2_{between}$ < 0.01. For sets to failure, there was a practically not worthwhile effect for shorter eccentric durations (g = 0.10 [-1.67, 1.87], (90% CI: [-1.21, 1.41], 95%PI = [-3.60, 3.81], 90%PI = [-2.64, 2.84], p = 0.87, I^2_{within} = 42.55%, $I^2_{between}$ = 42.55%, τ^2_{within} = 0.52,

$\tau^2_{between}$ = 0.52). Moderator analysis did not suggest differences between volume matched and sets to failure subgroups (β = 0.18 [-1.16, 1.52], 90% CI [-0.88, 1.24], F[1, 5.88] = 0.10, p = 0.76).

Hypertrophy

Pooled meta-analysis for hypertrophy identified a practically not worthwhile effect for faster eccentric duration (g = 0.02 [-0.44, 0.48], 90% CI: [-0.32, 0.37], 95%PI = [-0.63, 0.68], 90%PI = [-0.46, 0.51], p = 0.87, I^2_{within} = 0.6%, $I^2_{between}$ = 15.85%, τ^2_{within} < 0.01, $\tau^2_{between}$ = 0.02).

Sub-group analysis identified a practically not worthwhile effect for longer eccentrics for between- (g = 0.24 [-1.82, 2.30], 90% CI: [-1.15, 1.62], 95%PI = [-3.34, 3.84], 90%PI = [-2.22, 2.69], p = 0.66, I^2_{within} = 0%, $I^2_{between}$ = 69.63%, τ^2_{within} < 0.001, $\tau^2_{between}$ = 0.45) and within- (g = 0.007 [-1.66, 1.68], 90% CI: [-0.82, 0.83], 95%PI = [-3.41, 3.89], 90%PI = [-0.82, 0.83], p = 0.96, I^2_{within} = 4.96%, $I^2_{between}$ = 0, τ^2_{within} < 0.01, $\tau^2_{between}$ < 0.01) participant study designs. Moderator analysis did not suggest differences between between-participant and within-participant subgroups (β = -0.15 [-1.76, 1.45], 90% CI [-1.31, 1.01], F[1, 2.66] = 0.10, p = 0.77).

Sub-group analysis of training status identified a practically not worthwhile effect favouring shorter eccentric duration for untrained participants (g = -0.02 [-1.03, 0.98], 90% CI: [-0.67, 0.63], 95%PI = [-1.35, 1.30], 90%PI = [-0.46, 0.51], p = 0.91, I^2_{within} = 0%, $I^2_{between}$ = 19.28%, τ^2_{within} < 0.01, $\tau^2_{between}$ = 0.03). In trained participants, there was a small effect favouring shorter eccentric durations (g = 0.46 [-8.20, 9.12], 90% CI: [-3.84, 4.76], 95%PI = [-13.56, 14.47], 90%PI = [-6.50, 7.43], p = 0.62, I^2_{within} = 0%, $I^2_{between}$ = 87.59%, τ^2_{within} < 0.001, $\tau^2_{between}$ = 0.75). Moderator analysis did not suggest differences between trained and untrained subgroups (β = -0.33 [-3.01, 2.35], 90% CI [-2.18, 1.52], F[1, 2.15] = 0.25, p = 0.66).

Sub-group analysis of arm exercise selection identified a practically not worthwhile effect for longer eccentric durations (g = 0.29 [-1.40, 1.98], 90% CI: [-0.83, 1.41], 95%PI = [-2.56, 3.13], 90%PI = [-1.59, 2.17], p = 0.51, I^2_{within} = 0%, $I^2_{between}$ = 65.75%, τ^2_{within} < 0.001, $\tau^2_{between}$ = 0.23. For thigh exercise selection, there was a small effect for longer eccentric durations (g = -0.10 [-3.76, 3.56], 90% CI: [-1.92, 1.72], 95%PI = [-5.43, 5.23], 90%PI = [-2.75, 2.54], p = 0.78, I^2_{within} = 0%, $I^2_{between}$ = 44.46%, τ^2_{within} < 0.01, $\tau^2_{between}$ = 0.09). Moderator analysis did not suggest differences between arm and thigh subgroups (β = -0.36 [-2.00, 1.27], 90% CI [-1.53, 0.80], F[1, 2.45] = 0.65, p = 0.49).

Power

Pooled meta-analysis for power identified a moderate effect for shorter eccentric duration (g = -0.73 [-1.60, 0.15], 90% CI: [-1.34, -0.12], 95%PI = [-1.60, 0.15], 90%PI = [-1.31, -0.13], p = 0.07, I^2_{within} = 0%, $I^2_{between}$ = 0%, τ^2_{within} < 0.001, $\tau^2_{between}$ < 0.001).

Sensitivity with rho = 0.8

Strength

Pooled meta-analysis for maximal strength identified a practically not worthwhile effect favouring slower eccentric duration (g = 0.17 [-0.39, 0.74], 90% CI: [-0.28, 0.63], 95%PI = [-1.23, 1.59], 90%PI [-0.97, 1.32], p = 0.49, I^2_{within} = 0%, $I^2_{between}$ = 66.02%, τ^2_{within} < 0.01, $\tau^2_{between}$ = 0.30).

Sub-group analysis identified a practically not worthwhile effect for longer eccentrics for between- (g = 0.21 [-0.71, 1.13], 90% CI: [-0.51, 0.93], 95%PI = [-1.94, 2.35], 90%PI = [-1.91, 2.33], p = 0.59, I^2_{within} = 0%, $I^2_{between}$ = 73.52%, τ^2_{within} < 0.001, $\tau^2_{between}$ = 0.57) and within- (g = 0.11 [-1.16, 1.39], 90% CI: [-0.52, 0.74], 95%PI = [-1.16, 1.39], 90%PI = [-0.52, 0.74], p = 0.47, I^2_{within} = 0%, $I^2_{between}$ = 0%, τ^2_{within} < 0.01, $\tau^2_{between}$ < 0.01) participant study designs. Moderator analysis did not suggest differences between between-participant and within-participant subgroups (β = -0.10 [-1.74, 1.53], 90% CI [-1.20, 0.99], F[1, 1.95] = 0.08, p = 0.81).

Sub-group analysis of training status identified a practically not worthwhile effect favouring shorter eccentric duration for untrained participants (g = -0.03 [-1.68, 1.62], 90% CI: [-1.25, 1.19], 95%PI = [-3.51, 3.44], 90%PI = [-2.60, 2.54], p = 0.95, I^2_{within} = 0%, $I^2_{between}$ = 84.65%, τ^2_{within} < 0.01, $\tau^2_{between}$ = 0.92). In trained participants, there was a small effect favouring longer eccentric durations (g = 0.32 [-0.05, 0.71], 90% CI: [0.06, 0.59], 95%PI = [-0.04, 0.72], 90%PI = [0.07, 0.60], p = 0.06, I^2_{within} = 0%, $I^2_{between}$ = 0%, τ^2_{within} < 0.01, $\tau^2_{between}$ < 0.01). Moderator analysis did not suggest differences between trained and untrained subgroups (β = -0.42 [-1.63, 0.80], 90% CI [-1.38, 0.55], F[1, 5.87] = 0.71, p = 0.43).

Sub-group analysis of multi-joint exercise selection identified a practically not worthwhile effect for shorter eccentric durations (g = -0.12 [-1.45, 1.21], 90% CI: [-1.10, 0.86], 95%PI = [-2.77, 2.54], 90%PI = [-2.08, 1.84], p = 0.79, I^2_{within} = 0%, $I^2_{between}$ = 73.21%, τ^2_{within} < 0.001, $\tau^2_{between}$ = 0.52). For single joint exercise selection, there was a small effect for longer eccentric durations (g = 0.40 [-0.47, 1.27], 90% CI: [-0.22, 1.02], 95%PI = [-1.10, 1.90], 90%PI = [-0.69, 1.48], p = 0.23, I^2_{within} = 24.87%, $I^2_{between}$ = 24.87%, τ^2_{within} = 0.06, $\tau^2_{between}$ = 0.06). Moderator analysis did not suggest differences between multi-joint and single-joint subgroups (β = 0.56 [-0.65, 1.77], 90% CI [-0.40, 1.52], F[1, 5.86] = 1.28, p = 0.30).

Sub-group analysis of volume matched identified a small effect for longer eccentric durations (g = 0.24 [-0.03, 0.52], 90%CI = [0.03, 0.46], 95%PI = [-0.03, 0.52], 90%PI = [0.05, 0.44], p = 0.06, I^2_{within} = 0%, $I^2_{between}$ = 0%, τ^2_{within} < 0.01, $\tau^2_{between}$ < 0.01). For sets to failure, there was a practically not worthwhile effect for shorter eccentric durations (g = 0.10 [-1.67, 1.87], 90% CI: [-1.21, 1.41], 95%PI = [-3.60, 3.80], 90%PI = [-2.64, 2.84], p = 0.87, I^2_{within} = 42.55%, $I^2_{between}$ = 42.55%, τ^2_{within} = 0.52,

$\tau^2_{between}$ = 0.52). Moderator analysis did not suggest differences between volume matched and sets to failure subgroups (β = 0.17 [-1.17, 1.51], 90% CI [-0.89, 1.24], F[1, 5.89] = 0.10, p = 0.76).

Hypertrophy

Pooled meta-analysis for hypertrophy identified a practically not worthwhile effect for faster eccentric duration (g = 0.04 [-0.42, 0.50], 90% CI: [-0.29, 0.38], 95%PI = [-0.55, 0.63], 90%PI = [-0.39, 0.47], p = 0.79, I^2_{within} = 0%, $I^2_{between}$ = 10.39%, τ^2_{within} = 0.01, $\tau^2_{between}$ < 0.01).

Sub-group analysis identified a practically not worthwhile effect for longer eccentrics for between- (g = 0.25 [-1.82, 2.30], 90% CI: [-1.11, 1.62], 95%PI = [-3.27, 3.78], 90%PI = [-2.12, 2.64], p = 0.64, I^2_{within} = 0%, $I^2_{between}$ = 68.84%, τ^2_{within} < 0.001, $\tau^2_{between}$ = 0.45) and within- (g = 0.004 [-1.66, 1.66], 90% CI: [-0.82, 0.83], 95%PI = [-3.34, 3.83], 90%PI = [-1.22, 1.23], p = 0.98, I^2_{within} = 21.39%, $I^2_{between}$ = 0, τ^2_{within} = 0.02, $\tau^2_{between}$ < 0.01) participant study designs. Moderator analysis did not suggest differences between between-participant and within-participant subgroups (β = -0.17 [-1.72, 1.37], 90% CI [-1.30, 0.95], F[1, 2.73] = 0.14, p = 0.73).

Sub-group analysis of training status identified a practically not worthwhile effect favouring shorter eccentric duration for untrained participants (g = -0.002 [-0.91, 0.90], 90% CI: [-0.59, 0.59], 95%PI = [-1.11, 1.11], 90%PI = [-0.72, 0.71], p = 0.99, I^2_{within} = 11.52%, $I^2_{between}$ = 0%, τ^2_{within} = 0.01, $\tau^2_{between}$ < 0.01). In trained participants, there was a small effect favouring shorter eccentric durations (g = 0.46 [-8.20, 9.13], 90% CI: [-3.84, 4.77], 95%PI = [-13.52, 14.45], 90%PI = [-6.48, 7.41], p = 0.62, I^2_{within} = 0.13%, $I^2_{between}$ = 87.35%, τ^2_{within} < 0.001, $\tau^2_{between}$ = 0.75). Moderator analysis did not suggest differences between trained and untrained subgroups (β = -0.32 [-2.95, 2.31], 90% CI [-2.18, 1.52], F[1, 2.19] = 0.24, p = 0.67).

Sub-group analysis of arm exercise selection identified a practically not worthwhile effect for longer eccentric durations (g = 0.29 [-1.38, 1.97], 90% CI: [-0.81, 1.40], 95%PI = [-2.50, 3.09], 90%PI = [-1.56, 2.15], p = 0.50, I^2_{within} = 0.32%, $I^2_{between}$ = 65.73%, τ^2_{within} < 0.01, $\tau^2_{between}$ = 0.23. For thigh exercise selection, there was a small effect for longer eccentric durations (g = -0.08 [-3.68, 3.52], 90% CI: [-1.87, 1.71], 95%PI = [-4.97, 4.81], 90%PI = [-2.51, 2.35], p = 0.82, I^2_{within} = 9.16%, $I^2_{between}$ = 27.81%, τ^2_{within} = 0.02, $\tau^2_{between}$ = 0.05). Moderator analysis did not suggest differences between arm and thigh subgroups (β = -0.36 [-2.01, 1.30], 90% CI [-1.53, 0.82], F[1, 2.4] = 0.63, p = 0.50).

Power

Pooled meta-analysis for power identified a moderate effect for shorter eccentric duration (g = -0.74 [-1.67, 0.19], 90% CI: [-1.37, -0.11], 95%PI = [-1.67, 0.19], 90%PI = [-1.37, -0.11], p = 0.08, I^2_{within} = 0%, $I^2_{between}$ = 0%, τ^2_{within} < 0.01, $\tau^2_{between}$ < 0.01).